Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Thursday 19 February 2026

1.1 Daily Limit

Inverse Sine and Cosine Limit

Evaluate the limit:

$$\lim_{x\to 0} \frac{\sin^{-1}(x)\cdot(1-\cos(x^2))}{x^5}$$

Solution: We can evaluate this limit by expanding the functions in the numerator using their Maclaurin series approximations near $x = 0$.

First, we know that as $x \to 0$, the inverse sine function has the asymptotic behavior:

$$\sin^{-1}(x) \sim x$$

Second, for the cosine function, the standard expansion is $\cos(u) \approx 1 - \frac{u^2}{2}$. By substituting $u = x^2$, we obtain:

$$\cos(x^2) \approx 1 - \frac{(x^2)^2}{2} = 1 - \frac{x^4}{2}$$

Thus, the term $(1 - \cos(x^2))$ becomes:

$$1 - \cos(x^2) \sim \frac{x^4}{2}$$

Substituting these asymptotic equivalents back into the limit expression yields:

$$L = \lim_{x\to 0} \frac{(x)\cdot\left(\frac{x^4}{2}\right)}{x^5}$$

$$L = \lim_{x\to 0} \frac{\frac{1}{2}x^5}{x^5} = \frac{1}{2}$$

Friday 20 February 2026

2.1 Daily Limit

Limit of a Logarithmic Summation

Evaluate the limit:

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(\ln\left((n^2+k^2)^{1/n}\right)-\ln\left(n^{2/n}\right)\right)$$

Solution: Using the properties of logarithms, we can pull the exponents out to the front of the terms:

$$S_n=\sum_{k=1}^{n}\left(\frac{1}{n}\ln(n^2+k^2)-\frac{2}{n}\ln(n)\right)$$

We can factor out $\frac{1}{n}$ and rewrite $2\ln(n)$ as $\ln(n^2)$:

$$S_n=\frac{1}{n}\sum_{k=1}^{n}\left(\ln(n^2+k^2)-\ln(n^2)\right)$$

Using the quotient rule for logarithms, this simplifies to:

$$S_n=\frac{1}{n}\sum_{k=1}^{n}\ln\left(\frac{n^2+k^2}{n^2}\right)=\frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\left(\frac{k}{n}\right)^2\right)$$

Taking the limit as $n\to\infty$, this precisely defines a Riemann sum for the definite integral of $f(x)=\ln(1+x^2)$ from 0 to 1:

$$L=\int_0^1\ln(1+x^2)\,dx$$

We evaluate this integral using integration by parts. Let $u=\ln(1+x^2)$ and $dv=dx$. Then $du=\frac{2x}{1+x^2}\,dx$ and $v=x$:

$$\int_0^1\ln(1+x^2)\,dx=\Big[x\ln(1+x^2)\Big]_0^1-\int_0^1\frac{2x^2}{1+x^2}\,dx$$

$$L=\ln(2)-\int_0^1\left(2-\frac{2}{1+x^2}\right)dx$$

$$L=\ln(2)-\Big[2x-2\tan^{-1}(x)\Big]_0^1$$

$$L=\ln(2)-\left(2-2\left(\frac{\pi}{4}\right)\right)=\ln(2)-2+\frac{\pi}{2}$$

Saturday 21 February 2026

3.1 Daily Limit

Infinite Product of Radicals

Evaluate the limit:

$$\lim_{x\to\infty}\left(\sqrt[4]{2}\cdot\sqrt[9]{2}\cdot\sqrt[16]{2}\cdots\sqrt[x^2]{2}\right)$$

Solution: Let the product be denoted by $P(x)$. We can rewrite the radicals as fractional exponents:

$$P(x) = 2^{\frac{1}{4}}\cdot 2^{\frac{1}{9}}\cdot 2^{\frac{1}{16}}\cdots 2^{\frac{1}{x^2}}$$

Assuming x is an integer for the indexing (or taking the floor of x), we can express the product using summation notation in the exponent:

$$P(x) = 2^{\sum_{k=2}^{x}\frac{1}{k^2}}$$

Taking the limit as $x\to\infty$, we evaluate the infinite series in the exponent:

$$L = \lim_{x\to\infty} P(x) = 2^{\sum_{k=2}^{\infty}\frac{1}{k^2}}$$

We recognize this series from the famous Basel problem, which states that the sum of the reciprocals of all positive squares is:

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}$$

Our series starts at $k=2$, meaning we simply subtract the first term ($k=1$):

$$\sum_{k=2}^{\infty}\frac{1}{k^2} = \left(\sum_{k=1}^{\infty}\frac{1}{k^2}\right) - \frac{1}{1^2} = \frac{\pi^2}{6} - 1$$

Therefore, the limit evaluates to:

$$L = 2^{\frac{\pi^2}{6}-1}$$

Sunday 22 February 2026

4.1 Daily Limit

Square Root of Nested Trigonometric Functions Limit

Evaluate the limit:

$$\lim_{x\to 0^-}\left(\sqrt{\frac{\ln(\cos(\sin(x)))}{\sin(\tan(x))}}\right)$$

Solution: We will determine the asymptotic behavior of both the numerator and the denominator as $x \to 0$.

Numerator: As $x \to 0$, $\sin(x) \sim x$. The inner cosine function can be approximated using its Maclaurin series $\cos(u) \approx 1 - \frac{u^2}{2}$.

$$\cos(\sin(x)) \approx \cos(x) \approx 1 - \frac{x^2}{2}$$

Applying the natural logarithm, using $\ln(1-u) \approx -u$ as $u \to 0$:

$$\ln(\cos(\sin(x))) \approx \ln\left(1 - \frac{x^2}{2}\right) \approx -\frac{x^2}{2}$$

Denominator: For the inner tangent function, as $x \to 0$, $\tan(x) \sim x$. Thus, substituting this into the outer sine function:

$$\sin(\tan(x)) \approx \sin(x) \approx x$$

Evaluating the Ratio: We substitute these approximations back into the limit expression:

$$\frac{\ln(\cos(\sin(x)))}{\sin(\tan(x))} \approx \frac{-\frac{x^2}{2}}{x} = -\frac{x}{2}$$

Now, taking the limit as $x \to 0^-$ (meaning x is negative and approaching zero):

$$L = \lim_{x\to 0^-}\sqrt{-\frac{x}{2}}$$

Because x approaches zero from the negative side, $-x/2$ is positive and approaches 0^+.

$$L = \sqrt{0} = 0$$

Monday 23 February 2026

5.1 Daily Limit

Riemann Sum with Trigonometric Functions

Evaluate the limit:

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{\pi}{2}+\frac{\pi k}{4n}\right)\cos\left(\frac{\pi}{2}+\frac{\pi k}{4n}\right)\left(\frac{\pi}{4n}\right)$$

Solution: This limit clearly describes a Riemann sum of a continuous function over a specific interval. Let $x_k = \frac{\pi}{2}+\frac{\pi k}{4n}$. The width of each subinterval is $\Delta x = \frac{\pi}{4n}$. As k goes from 1 to n, the value of x_k traverses from $\frac{\pi}{2}$ to $\frac{\pi}{2}+\frac{\pi n}{4n} = \frac{\pi}{2}+\frac{\pi}{4} = \frac{3\pi}{4}$. The limit of the sum translates to the definite integral:

$$L = \int_{\pi/2}^{3\pi/4} x\cos(x)\,dx$$

We solve this integral using integration by parts. Let $u = x$ and $dv = \cos(x)\,dx$. Then $du = dx$ and $v = \sin(x)$:

$$\int x\cos(x)\,dx = x\sin(x) - \int \sin(x)\,dx = x\sin(x)+\cos(x)$$

Evaluating this from $\frac{\pi}{2}$ to $\frac{3\pi}{4}$:

$$\begin{aligned}
L &= \left[\frac{3\pi}{4}\sin\left(\frac{3\pi}{4}\right)+\cos\left(\frac{3\pi}{4}\right)\right]-\left[\frac{\pi}{2}\sin\left(\frac{\pi}{2}\right)+\cos\left(\frac{\pi}{2}\right)\right] \\
&= \left(\frac{3\pi}{4}\cdot\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}\right)-\left(\frac{\pi}{2}\cdot 1+0\right) \\
&= \frac{3\pi\sqrt{2}}{8}-\frac{\sqrt{2}}{2}-\frac{\pi}{2}
\end{aligned}$$

Tuesday 24 February 2026

6.1 Daily Limit

Geometric Series Defined by a Limit

Evaluate the limit:

$$\lim_{x\to 0}\left(\sum_{k=0}^{1/x}\left(\frac{e^{-xe}\tan^{-1}\left(\frac{1}{x}\right)\sin\left(\frac{\pi}{3}\right)}{2\cos(x)(1+x)^{\frac{1}{x}}}\right)^k\right)$$

Solution: Let us assume $x \to 0^+$. Under this assumption, the upper limit of the summation $N = \frac{1}{x} \to \infty$. This transforms the expression into an infinite geometric series. Let the base of the geometric term be $R(x)$. We first find $r = \lim_{x\to 0^+} R(x)$:

$$R(x) = \frac{e^{-xe}\tan^{-1}\left(\frac{1}{x}\right)\sin\left(\frac{\pi}{3}\right)}{2\cos(x)(1+x)^{\frac{1}{x}}}$$

Evaluating the limits of the individual components as $x \to 0^+$:

- $e^{-xe} \to e^0 = 1$
- $\tan^{-1}\left(\frac{1}{x}\right) \to \frac{\pi}{2}$
- $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$
- $\cos(x) \to 1$
- $(1+x)^{1/x} \to e$

Substituting these values, the base converges to a constant ratio r:

$$r = \frac{1\cdot\left(\frac{\pi}{2}\right)\cdot\left(\frac{\sqrt{3}}{2}\right)}{2\cdot(1)\cdot(e)} = \frac{\frac{\pi\sqrt{3}}{4}}{2e} = \frac{\pi\sqrt{3}}{8e}$$

Since $8e \approx 21.75$ and $\pi\sqrt{3} \approx 5.44$, the ratio $|r| < 1$. As $x \to 0^+$, the number of terms $\frac{1}{x}$ goes to infinity, and the series becomes a standard convergent infinite geometric series $\sum_{k=0}^{\infty} r^k$.

$$L = \frac{1}{1-r} = \frac{1}{1-\frac{\pi\sqrt{3}}{8e}} = \frac{8e}{8e-\pi\sqrt{3}}$$

Wednesday 25 February 2026

7.1 Daily Limit

Factorial Dominance Limit

Evaluate the limit:
$$\lim_{x \to \infty} \frac{\sqrt{(x!)^2 + 1}}{x!}$$

Solution: To evaluate this limit as $x \to \infty$, we can factor out the dominant term, $(x!)^2$, from underneath the square root:

$$L = \lim_{x \to \infty} \frac{\sqrt{(x!)^2 \left(1 + \frac{1}{(x!)^2}\right)}}{x!}$$

Assuming $x \geq 0$, we can simplify the square root, since $\sqrt{(x!)^2} = x!$. This gives:

$$L = \lim_{x \to \infty} \frac{x! \sqrt{1 + \frac{1}{(x!)^2}}}{x!}$$

We can now cancel the $x!$ term from both the numerator and the denominator:

$$L = \lim_{x \to \infty} \sqrt{1 + \frac{1}{(x!)^2}}$$

As $x \to \infty$, the factorial function $x!$ grows to infinity, which means the fraction $\frac{1}{(x!)^2}$ rapidly approaches 0. Substituting this into the expression yields:

$$L = \sqrt{1 + 0} = 1$$

Thursday 26 February 2026

8.1 Daily Limit

Complex Exponential Limit

Evaluate the limit:

$$\lim_{x\to\infty}\left(1+\frac{i\pi}{x}\right)^x$$

Solution: This limit is a classic application of the definition of the exponential function. The limit representation of e^z for any complex number z is given by:

$$e^z = \lim_{x\to\infty}\left(1+\frac{z}{x}\right)^x$$

In our given problem, we can identify $z = i\pi$. Substituting this into the standard exponential limit formula, we get:

$$L = e^{i\pi}$$

To evaluate $e^{i\pi}$, we invoke Euler's Identity, which bridges trigonometry and complex exponentials: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$.

$$L = \cos(\pi) + i\sin(\pi) = -1 + i(0) = -1$$

Friday 27 February 2026

9.1 Daily Limit

Factorial Ratio with Exponentials Limit

Evaluate the limit:

$$\lim_{x\to\infty} \frac{(x!)^2 \cdot 4^x}{(2x)! \cdot \sqrt{\pi x}}$$

Solution: To resolve limits involving large factorials, we employ Stirling's approximation, which states $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$.

First, we approximate the terms in the numerator:

$$x! \sim \sqrt{2\pi x}\left(\frac{x}{e}\right)^x$$

Squaring this yields:

$$(x!)^2 \sim \left(\sqrt{2\pi x}\left(\frac{x}{e}\right)^x\right)^2 = 2\pi x\left(\frac{x}{e}\right)^{2x}$$

Next, we approximate $(2x)!$ in the denominator:

$$(2x)! \sim \sqrt{2\pi(2x)}\left(\frac{2x}{e}\right)^{2x} = \sqrt{4\pi x}\frac{2^{2x}x^{2x}}{e^{2x}} = 2\sqrt{\pi x}\cdot 4^x\left(\frac{x}{e}\right)^{2x}$$

Now, substitute these approximations back into the original limit expression:

$$L = \lim_{x\to\infty} \frac{\left[2\pi x\left(\frac{x}{e}\right)^{2x}\right]\cdot 4^x}{\left[2\sqrt{\pi x}\cdot 4^x\left(\frac{x}{e}\right)^{2x}\right]\cdot\sqrt{\pi x}}$$

We can simplify the denominator by multiplying the square roots together:

$$(2\sqrt{\pi x})\cdot(\sqrt{\pi x}) = 2\pi x$$

The denominator simplifies entirely to:

$$2\pi x\cdot 4^x\left(\frac{x}{e}\right)^{2x}$$

Since the numerator and the denominator are asymptotically identical, their ratio converges exactly to 1.

$$L = \lim_{x\to\infty} \frac{2\pi x\left(\frac{x}{e}\right)^{2x}4^x}{2\pi x\left(\frac{x}{e}\right)^{2x}4^x} = 1$$

Saturday 28 February 2026

10.1 Daily Limit

Sine Limits with Squeeze Theorem

Evaluate the limit:

$$\lim_{x\to 0}\left[x^{2026}\cdot\sin\left(\frac{2026}{x^{2026}}\right)-\sin\left(x^{2026}\right)\cdot\frac{2026}{x^{2026}}\right]$$

Solution: We can evaluate this limit by splitting it into two separate limit evaluations:

$$L_1=\lim_{x\to 0}x^{2026}\sin\left(\frac{2026}{x^{2026}}\right)$$

$$L_2=\lim_{x\to 0}\sin\left(x^{2026}\right)\cdot\frac{2026}{x^{2026}}$$

First Term (L_1): The sine function is bounded between -1 and 1. Therefore, for all $x\neq 0$:

$$-1\leq\sin\left(\frac{2026}{x^{2026}}\right)\leq 1$$

Multiplying by x^{2026} (which is always non-negative since the exponent is even):

$$-x^{2026}\leq x^{2026}\sin\left(\frac{2026}{x^{2026}}\right)\leq x^{2026}$$

Taking the limit as $x\to 0$ for the bounds gives 0 on both sides. By the Squeeze Theorem:

$$L_1=0$$

Second Term (L_2): Rearranging L_2, we have:

$$L_2=2026\lim_{x\to 0}\frac{\sin\left(x^{2026}\right)}{x^{2026}}$$

Let $u=x^{2026}$. As $x\to 0$, $u\to 0$. This gives the fundamental trigonometric limit form:

$$L_2=2026\lim_{u\to 0}\frac{\sin(u)}{u}=2026\cdot(1)=2026$$

Final Limit: Subtracting the evaluations of the two separated limits:

$$L=L_1-L_2=0-2026=-2026$$